Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN STACK

Practical 2

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Class: TYIT	Batch: 1
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AIM :- Core Javascripts Concepts for MERN

A) 1. Create functions and call them with parameters.

CODE :-

```
function calculateSalary(name,basicSalary) {

    let bonus = 5000;

    let totalSalary = basicSalary + bonus;

    console.log("Employee Name : ",name);

    console.log("Basic Salary : ",basicSalary);

    console.log("Bonus : ",bonus);

    console.log("Total Salary : ",totalSalary);

}

calculateSalary("Ritesh Chaubey",35000);
```

OUTPUT :-

```
Employee Name :  Ritesh Chaubey
Basic Salary :  35000
Bonus :  5000
Total Salary :  40000
```

2. **CODE :-**

```
function generateBill(product, quantity, price)
```

```
{
    let total = quantity * price;
    let bill = total + (total * 0.09);
    console.log("Product : ", product);
    console.log("Quantity : ", quantity);
    console.log("Price Per Item : ", price);
    console.log("Total Bill : ", bill);
}
generateBill("Laptop",2,50000);
```

OUTPUT :-

```
PS C:\Users\mvluc\Desktop\MERN Practical> node Prac2A.js
Product :  Laptop
Quantity :  2
Price Per Item :  50000
Total Bill :  109000
```

B) Works with Arrays and Object in Javascripts

CODE :-

```
let fruits = ["Apple","Banana","Guava"];
let Student = {
    Name :"Ritesh Chaubey",
    Age : 20,
    City : "Mumbai"
};
console.log("Fruits : ", fruits);
console.log("Second Fruit : ", fruits[1]);

console.log("Student Name : ", Student.Name);
```

```
console.log("Student Age : ", Student.Age);

console.log("Student City : ", Student.City);
```

OUTPUT :-

```
Fruits :  [ 'Apple', 'Banana', 'Guava' ]
Second Fruit :  Banana
Student Name :  Ritesh Chaubey
Student Age :  20
Student City :  Mumbai
```

C) Demonstrate use of arrow functions.

CODE :-

```
let add = (a,b) => {

    return a + b;

};

let sub = (a,b) => {

    return a - b;

};

let mul = (a,b) => {

    return a * b;

};

let div = (a,b) => {

    return a / b;

};

console.log("Addition = ", add(10, 20));

console.log("Subtraction = ", sub(10, 20));

console.log("Multiplication = ", mul(10, 20));

console.log("Division = ", div(10, 20));
```

OUTPUT :-

```
PS C:\Users\mvluc\Desktop\MERN Practical> node Prac2C.js
Addition =  30
Subtraction =  -10
Multiplication =  200
Division =  0.5
```

D) Write a program using array methods (map, filter, reduce).

CODE :-

```
let numbers = [10,20,30,40,50];

let doubled = numbers.map(num => num * 2);

let greater25 = numbers.filter(num => num > 25);

let total = numbers.reduce((sum,num) => sum + num, 0);


console.log("Original Array : ",numbers);

console.log("Map : ",doubled);

console.log("Filter : ",greater25);

console.log("Reduce : ",total);
```

OUTPUT :-

```
PS C:\Users\mvluc\Desktop\MERN Practical> node Prac2D.js
Original Array :  [ 10, 20, 30, 40, 50 ]
Map :  [ 20, 40, 60, 80, 100 ]
Filter :  [ 30, 40, 50 ]
Reduce :  150
```

E) Create a program that manipulates objects and displays output.

CODE :-

```
let student = {

    rollNo: 13,

    Name: "Ritesh Chaubey",

    Marks: 67
```

```
};


console.log("\n Accessing Properties: ");

console.log("Name : ", student.Name);

console.log("Marks : ", student.Marks);


// Updating Property

student.Marks = 69;

console.log("\n After Updating Marks : ");

console.log(student);


// Adding a Property

student.city = "Mumbai";

console.log("\nAfter Adding City : ");

console.log(student);


// Deleting

delete student.rollNo;

console.log("\nAfter Deleting : ");

console.log(student);
```

OUTPUT :-

```
 Accessing Properties:
Name :  Ritesh Chaubey
Marks :  67

 After Updating Marks :
{ rollNo: 13, Name: 'Ritesh Chaubey', Marks: 69 }

After Adding City :
{ rollNo: 13, Name: 'Ritesh Chaubey', Marks: 69, city: 'Mumbai' }

After Deleting :
{ Name: 'Ritesh Chaubey', Marks: 69, city: 'Mumbai' }
```

F) Implement a small JavaScript program combining functions, arrays, and objects.

CODE :-

```
let students = [

    {

    rollNo: 13,

    Name: "Saqlain Khan",

    Marks: 67

},

{

    rollNo: 14,

    Name: "Ritesh",

    Marks: 68

},

{

    rollNo: 15,

    Name: "Himani",

    Marks: 69

}
```

```
];

function display(s)

{

    console.log("Student Details");

    console.log("-----------------");

    for (let stud of s)

{

    console.log(`Roll No : ${stud.rollNo}`);

    console.log(`Name : ${stud.Name}`);

    console.log(`Marks : ${stud.Marks}`);

    console.log(`--------------------------`);

}

}

display(students);
```

OUTPUT :-

```
--------------------------
PS C:\Users\mvluc\Desktop\MERN Practical> node Prac2F.js
Student Details
----------------
Roll No : 13
Name : Saqlain Khan
Marks : 67
------------------------
Roll No : 14
Name : Ritesh
Marks : 68
------------------------
Roll No : 15
Name : Himani
Marks : 69
```